PIDray Journal Method: Deep Theoretical and Experimental Review

This document focuses only on the scientific contribution of the PIDray journal paper: the problem formulation, dataset design, model reasoning, losses, experimental protocol, published results, ablations, and limitations. Repository structure and implementation commands are intentionally separated into the companion repository document.

1. Research Problem and Journal Contribution

Real security-inspection streams contain far more normal baggage images than images with prohibited items. A detector trained mainly on positives can learn to force foreground predictions on ordinary contents, while hidden or overlapping items remain difficult because X-ray appearance differs from natural images. PIDray addresses these conditions with a large real-world benchmark and a divide-and-conquer task formulation.

The journal extension contributes:

- a 124,486-image dataset with 76,809 normal images and 47,677 positive images from multiple locations and scanner manufacturers;
- bounding boxes, instance masks, and image-level multi-label annotations for 12 prohibited-item categories;
- Easy, Hard, and Hidden test subsets that represent different operational difficulties;
- a coarse image-level node R_0 that estimates whether a bag contains any prohibited item;
- a fine node R_1 based on Cascade Mask R-CNN, strengthened by dense cross-scale attention;
- experiments on detection, instance segmentation, multi-label classification, general detectors, and ablations.

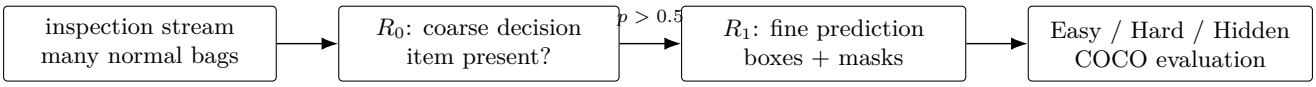


Figure 1: The paper's divide-and-conquer reasoning, independently redrawn.

2. Dataset Design and Evaluation Questions

Table 1: PIDray image composition reported in the journal paper.

Partition	Images	Positive	Normal	Scientific purpose
Full dataset	124,486	47,677	76,809	Realistic positive/normal imbalance; 12 categories.
Train	76,913	29,454	47,459	Training population.
Test–Easy	24,758	9,482	15,276	One exposed item in each positive image.
Test–Hard	9,746	3,733	6,013	Multiple items; stronger overlap and clutter.
Test–Hidden	13,069	5,008	8,061	Deliberately concealed or distorted item.
Test total	47,573	18,223	29,350	Combined operational evaluation.

The three subsets answer different questions. Easy measures recognition/localization under relatively clean conditions. Hard measures multiple-instance separation and overlap. Hidden measures robustness to deliberate concealment, where proposal recall and feature discrimination become critical. Overall AP must therefore be accompanied by subset metrics; otherwise a method can improve common images while degrading the most security-sensitive cases.

3. Theoretical Model

3.1 Backbone and five-level feature pyramid

The fine detector begins with ResNet-101 and a feature pyramid. Backbone outputs C_2–C_5 have strides 4, 8, 16, and 32. A fifth pooled pyramid level produces stride 64. Every pyramid output has 256 channels. Standard top-down FPN fusion is followed by dense attention that lets each target level use information from all five source levels.

For target level ℓ, source feature P_j is resized to the target grid:

$$\widetilde{P}_{j\to\ell} = \begin{cases} \text{AdaptiveMaxPool}(P_j), & j < \ell, \\ \text{NearestResize}(P_j), & j \geq \ell. \end{cases}$$

The resulting tensor has shape $B \times 5 \times 256 \times H_\ell \times W_\ell$.

3.2 Channel and spatial selection

Channel attention estimates how useful each source level is for each channel. The five resized features are summed, globally average-pooled, projected through a $256 \to 128$ bottleneck, then expanded through five $128 \to 256$ projections. A softmax across levels produces $a^c_{\ell,j}$.

Spatial attention first sums the levels, then concatenates channel-wise mean and maximum maps. Five 7×7 convolutions and a level-wise softmax produce spatial weights $a^s_{\ell,j}(u, v)$. The two branches form

$$U^c_\ell = \sum_{j=1}^{5} a^c_{\ell,j} \odot \widetilde{P}_{j\to\ell}, \qquad U^s_\ell = \sum_{j=1}^{5} a^s_{\ell,j} \odot \widetilde{P}_{j\to\ell}.$$

A global-context block refines their sum, followed by a residual connection:

$$P'_\ell = P_\ell + \mathrm{ContextBlock}_{256,1/16}(U^c_\ell + U^s_\ell).$$

This design tries to preserve small-item detail while importing semantic evidence from coarser levels and long-range context from the refinement block.

3.3 Coarse node R_0

The coarse node receives the final refined pyramid level, a normalized 2-D sine positional encoding, and one learned 256-dimensional query. A two-layer transformer decoder produces one logit z_i per image. The target y_i is 1 if any ground-truth box exists and 0 otherwise:

$$L_{R_0} = 0.1\ \mathrm{BCEWithLogits}(z, y), \qquad p_i = \sigma(z_i).$$

The paper routes images with $p_i > 0.5$ to the fine detector and scales the fine loss by the selected/positive proportion to stabilize learning when few samples enter R_1. Conceptually, R_0 separates the common question *is any threat present?* from the expensive questions *where, what class, and what mask?*

3.4 Fine node R_1

R_1 is a three-stage Cascade Mask R-CNN. RPN proposes class-agnostic regions. Three box stages use increasing IoU thresholds 0.5, 0.6, and 0.7, encouraging progressively higher-quality localization. A class-specific FCN mask head predicts one mask per positive region. The detection/segmentation objective can be summarized as

$$L = L_{R_0} + \lambda_{\mathrm{route}} \left(L_{\mathrm{rpn_cls}} + L_{\mathrm{rpn_box}} + \sum_{t=1}^{3} w_t (L^{(t)}_{\mathrm{cls}} + L^{(t)}_{\mathrm{box}}) L_{\mathrm{mask}} \right),$$

where $(w_1, w_2, w_3) = (1, 0.5, 0.25)$ and λ_{route} is the batch proportion used to scale fine-node gradients.

4. Experimental Protocol

Table 2: Paper-level detection and segmentation protocol.

Element	Reported protocol
Base detector	Cascade Mask R-CNN with ResNet-101 and the proposed pyramid/gate.
Input resolution	Approximately 500×500, matching the dataset's average resolution.
Optimization	SGD, initial LR 0.02, momentum 0.9, weight decay 10^{-4}, batch 16.
Training length	12 epochs for two-stage models; 24 for one-stage comparisons.
Hardware	Four NVIDIA RTX 3090 GPUs for detection/segmentation.
Metrics	COCO AP/AR over IoU 0.50:0.05:0.95; box and segmentation results.
Primary stratification	Easy, Hard, Hidden, and combined test sets.

The journal comparison includes one-stage and two-stage detectors: SSD variants, FCOS, RetinaNet, YOLOv3, YOLOX-M, TOOD, DW, DDOD, Faster R-CNN, Libra R-CNN, Mask R-CNN, Cascade R-CNN, Cascade Mask R-CNN, a PAFPN variant, and the earlier SDANet. This breadth tests whether gains arise only from one detector family.

5. Published Detection and Segmentation Results

Table 3: Main published PIDray results.

	Box AP				Mask AP			
Method	Easy	Hard	Hidden	All	Easy	Hard	Hidden	All
Mask R-CNN	66.2	58.6	43.8	59.1	59.2	50.1	35.5	51.2
Cascade Mask R-CNN	71.9	63.2	46.8	63.7	60.7	52.0	36.2	52.8
SDANet	72.5	63.7	48.0	64.4	61.1	51.7	37.0	52.9
Journal method	74.5	64.8	53.0	66.6	61.4	51.9	39.7	53.4

Against plain Cascade Mask R-CNN, the journal method gains 2.9 overall box AP and 6.2 Hidden box AP, but only 0.6 overall mask AP. The strongest evidence is therefore improved localization, especially for concealed items. The segmentation improvement is positive but modest overall.

6. Ablations and Generalization Evidence

Table 4: Selected published mechanism ablations.

Comparison	**Box AP**	**Mask AP**	**Inference**
DDOD → DDOD + divide/conquer	63.6 → 64.5	—	Coarse screening transfers to one-stage detection.
Cascade Mask R-CNN → + gate	63.7 → 65.4	52.5 → 52.7	Gate primarily improves box detection.
+ channel attention	65.4 → 65.9	52.7 → 53.2	Channel selection adds complementary evidence.
+ spatial attention alone	65.4 → 65.7	52.7 → 52.7	Spatial branch alone helps boxes.
Full CA + SA + dependency refinement	66.6	53.4	Complete dense attention gives final performance.

Normal-image ablation is central to the study's background-confusion objective. Excluding normal images gives 63.2 detection AP and 21.4% image-level error; the full dataset gives 63.7 AP and 4.8% error. Normal images therefore contribute little AP but dramatically improve the decision not to raise an alarm.

Table 5: Published AP before and after adding R_0.

Method	**AP**	**+R0**	**Method**	**AP**	**+R0**
FCOS	54.1	57.0	DDOD	63.6	64.5
RetinaNet	57.6	58.6	Faster R-CNN	58.3	58.4
YOLOX-M	61.3	61.5	Libra R-CNN	56.7	57.6
TOOD	62.2	64.0	Mask R-CNN	59.1	59.7
DW	61.6	63.7	Cascade Mask R-CNN	63.7	65.4

7. Scientific Interpretation and Limitations

Table 6: What the experiments do and do not establish.

Evidence	Responsible interpretation
Large Hidden box gain	Dense cross-scale features and screening improve concealed-item localization.
Small overall mask gain	The method is not primarily an occlusion-specific segmentation solution.
Normal-image error reduction	Negative images are necessary for realistic alarm behavior.
R0 improves many detectors	The coarse/fine decomposition is general, but gain size varies strongly by detector.
Single published run per setting	Statistical uncertainty and seed sensitivity are not quantified.
Test subsets used for final reporting	New research must create train-derived validation rather than tune on them.
Gate before fine reasoning	Gate false negatives can erase otherwise detectable positive images and must be reported separately.

For the proposed study, PIDray is the primary baseline for background confusion at image level. A new component should target proposal-level or instance-level confusion, mask leakage, or difficult-positive routing, and should be justified by an error taxonomy rather than duplicating the existing gate.

8. Primary Source

Zhang, L., Jiang, L., Ji, R., and Fan, H. (2023). *PIDray: A Large-Scale X-ray Benchmark for Real-World Prohibited Item Detection*. International Journal of Computer Vision, 131, 3170–3192. arXiv; journal DOI.